

Khanh Nguyen (Alex) Bui

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OBJECTIVE: A dedicated Business Analytics and Data Science student with strong analytical thinking and problem-solving skills. Seeking opportunities to apply statistical modeling, predictive analytics, and data-driven insights to business challenges.

Education

Bachelor of Science (Honours) - Data Science (Expected: July 2027) July 2026 - July 2027
The University of Sydney

• **Honours Supervisor:** Dr. Richard Gerlach

Bachelor of Commerce (Grade: **Distinction**) Feb 2024 - July 2026
The University of Sydney

- **Major:** Business Analytics and Data Science
- **Relevant Topics:** Statistical Modeling, Integer/Non-Integer Optimization, Dynamic Programming, Stochastic Processes, Bayesian Inference, Generalized Linear Models, Time Series Analysis, Option Pricing, Machine Learning.

Skills & Interest

Technical Skills: Python, R, SQL, LaTeX, Quarto, Git.
Academic Interest: Time Series Analysis, Portfolio Optimization, Machine Learning, Option Pricing.
Other Interests: Golf, Photography, Poker.

Experience

Receptionist May 2025 - Sep 2025
BeBee Co-Working Space - HCMC, Vietnam

- Met and greeted up to 200+ guests daily to ensure they received a warm and friendly welcome.
- Took time to understand the needs of each visitor and offered additional information.
- Kept accurate records on check-ins and maintained data confidentiality.
- Worked as part of a team of 5 and motivated others whenever possible.

Waiter May 2024 - Sep 2024
Ngon 138 Restaurant - HCMC, Vietnam

- Served customers and addressed complaints to improve customer service.
- Developed verbal communication skills by engaging with manager and kitchen staff to stay up-to-date on supply availability.
- Developed time management skills to minimize customer wait times.

Full-time Junior Golfer Jun 2019 - Dec 2021
MMG Performance - Orlando, Florida

- Trained with world-class coaches and competed in various junior tournaments.
- Achieved the best personal ranking of 7th in Disney Summer Junior Open (+10 over two rounds).
- Developed self-discipline through training and mental toughness through competition.

Technical Projects

2-Regime Portfolio Strategy Comparison  Dec, 2025

- Implemented and compared three regime-switching portfolio strategies using volatility hysteresis and trend detection on 12 liquid ETFs spanning sectors, bonds, and gold.
- Applied Ledoit-Wolf shrinkage for robust covariance estimation and optimized portfolios using Max Return, Max Sharpe, and Min CVaR objectives.
- Accelerated computationally intensive loops using Numba JIT compilation, achieving 10-50x speedups.

- Achieved superior risk-adjusted performance: Strategy 2 (Sharpe/CVaR) delivered the highest Sharpe ratio with significantly reduced maximum drawdown vs. SPY Buy&Hold.
- **Tools Used:** **Python** (NumPy, SciPy, Numba, Scikit-learn, Plotly, yfinance), Quarto
- **Technical Skills:** Portfolio Optimization, CVaR, Regime Detection, JIT Compilation, Shrinkage Estimation.

Regime-Switching Volatility Forecasting

Dec, 2025

- Fetched market data using Quantmod and wrote scripts to resolve timezone-related date errors.
- Applied Hidden Markov Models within a GJR-GARCH framework to forecast volatility and identify distinct market stress regimes.
- Selected Skewed Student-t distributions to accurately model 'fat tails' and asymmetry in market data.
- Reduced MSE by 1.35% against single-regime benchmarks via parallelized expanding window cross-validation.
- **Tools Used:** **R** (MSGARCH, Quantmod, Tidyverse, doParallel, Plotly), Quarto
- **Technical Skills:** Volatility Modeling, Hidden Markov Models, Parallel Computing.

Bayesian Implementation: MCMC vs Variational Bayes for ASX200

Nov, 2025

- Implemented Random Walk Metropolis and Variational Bayes algorithms from scratch to estimate log-STD-GARCH(1,1) volatility parameters.
- Optimized VB using reparameterisation gradients, Numba JIT compilation, and parallel Monte Carlo to achieve competitive runtime with MCMC.
- **Tools Used:** **Python** (NumPy, Numba, SciPy, Joblib, Matplotlib), Quarto
- **Technical Skills:** Bayesian Inference, MCMC, Variational Bayes, JIT Optimization, Parallel Computing.

Dietary Nutrients and Blood Pressure: A Statistical Analysis

Nov, 2025

- Cleaned Australian Health Survey data using Goldberg cutoffs to exclude implausible energy intakes, reducing sample from 12,000+ to 4,250 valid observations.
- Constructed DAGs to identify confounders and fitted bootstrapped ridge regression to handle multicollinearity among correlated nutrients.
- **Tools Used:** **R** (Tidyverse, glmnet, boot, ggdag, gt), Quarto
- **Technical Skills:** Causal Inference (DAGs), Ridge Regression, Bootstrap Inference, Data Cleaning.

Australian CPI Forecasting

May, 2025

- Developed an ARIMA forecasting pipeline for Australian CPI data using Python, Statsmodels, and Pandas.
- Implemented exhaustive grid search with expanding window cross-validation to identify optimal hyperparameters.
- Surpassed baseline model performance by reducing Mean Squared Error (MSE) through out-of-sample validation.
- **Tools Used:** **Python** (Statsmodels, Pandas, Scikit-learn, Matplotlib, Seaborn), Quarto, GitHub Pages
- **Technical Skills:** Time Series Modeling, Expanding Window Cross-Validation, Data Visualization.